PORTFOLIO OF SELECTED WORKS

# Adam E. Kaminsky

SHEEP I MANAGED AT
ROCKY CREEK FARM, MT

TEACHING WINTER ECOLOGY ON SKIS, CROSSCUT MOUNTAIN SPORTS CENTER, MT (PHOTO CREDIT: CROSSCUT STAFF)

# Adam E. Kaminsky

802-735-5894
minsky@umich.edu

I grew up exploring the forests, swamps, and farms around my childhood home in rural Vermont. My passion for the natural world led me to pursue a life outdoors, and develop a participatory relationship with nature. My diverse experience engaging directly with the land forms the backbone of my work, which seeks to promote eco-cultural revitalization in response to global challenges.

## Education

**University of Michigan**
Master of Landscape Architecture
Master of Science: *Ecosystem Science and Management and Geospatial Data Science*

**Middlebury College**
Bachelor of Arts: *Conservation Biology (major) and Studio Art (minor)*

**Universidad Austral de Chile**
Study abroad with coursework in restoration, forestry, and Spanish language

## Experience

**Wyss Scholar | Quivira Coalition |** *New Mexico*
Project lead on Rio Puerco watershed restoration report

**Ecological Restoration Technician | Self-Employed |** *Montana, New Mexico, Vermont*
Ecological restoration on working lands focusing on hydrological restoration, erosion control, biodiversity, and livestock management

**Education Director | Rocky Creek Farm |** *Montana*
Founder, director and lead instructor of education programming on diversified organic farm

**Diversified Organic Farming and Ranching | Various |** *Vermont, Montana, Michigan*
Extensive experience with organic techniques for farming vegetables, livestock, and flowers

**Winter Ecology Ski Instructor | Crosscut Mountain Sports Center |** *Montana*
Outdoor education and ski instruction for kids of all ages, and naturalist classes on skis for adults

**Youth Crew Leader | Montana Conservation Corps |** *Montana*
Crew leader for high-school-aged kids working in backcountry conservation, restoration, and recreation projects in the Greater Yellowstone Ecoregion

## Skills

Education, Facilitation and Negotiation

Farming *(vegetables, flowers, cover crops, grains)*

Livestock and Poultry Management and Ranching *(Cattle, sheep, chickens, turkeys, pigs, regenerative grazing)*

Ecological Restoration (*Planting, Erosion Control, Rock Work, Log Work, Fire Mitigation, Prescribed Burns, LTPBR*)

GIS and Remote Sensing (*ESRI Suite, QGIS, Satellite Imagery, Drone flights and data analysis*)

Publication and Graphic Design, 2D / 3D Rendering, Drafting (*Adobe Suite, Affinity Suite, Rhino, CAD, hand rendering*)

Backcountry Travel / Navigation (*On foot, bike, ski, canoe, etc.*)

## Portfolio Contents

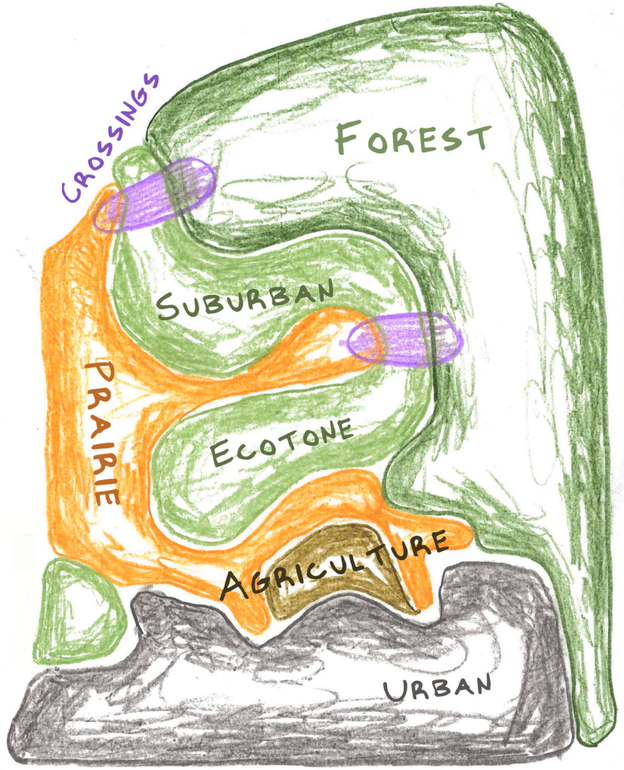
*Site Design*
*Trail Networks*
*Conservation Design*
*Restoration Design*
*Construction Drawings and Grading*
*Non-Profit Work*
*Geospatial Data Analysis*
*Suitability Analysis*
*Restoration Work*

CREEK RESTORATION PROJECT IN ABSAROKEE, MT

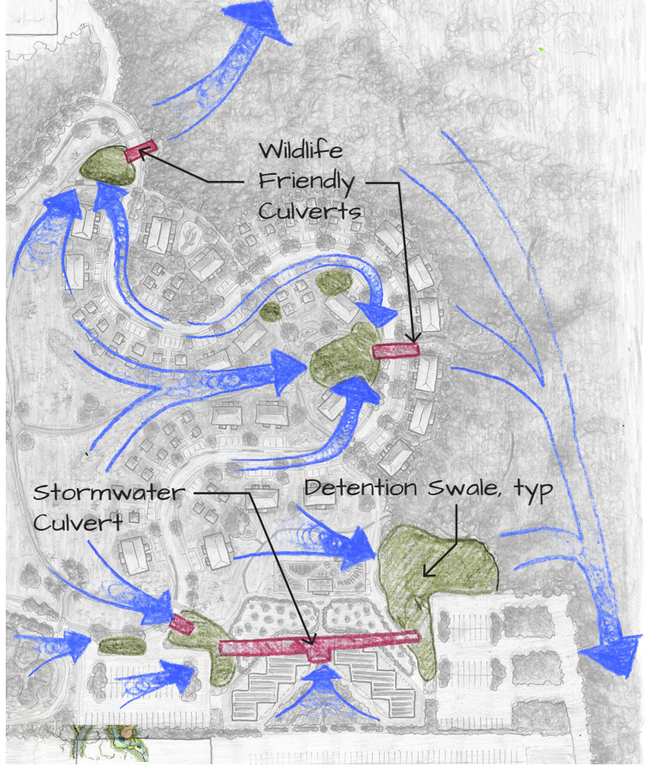
# **Site Design** | Suburban Ecotone Development | Ann Arbor, MI

Ecotones are the unique spaces between distinct ecosystem types, forming an interstitial space of connection. In this design, humans are placed at the ecotone between restored forest and prairie to promote relationships between humans and nature. Inspired by biological strategies to maximize surface area, the sinuous design maximizes housing, while simultaneously minimizing fragmentation and ensuring that every household has direct access to natural spaces.

## Landscape Matrix Diagram



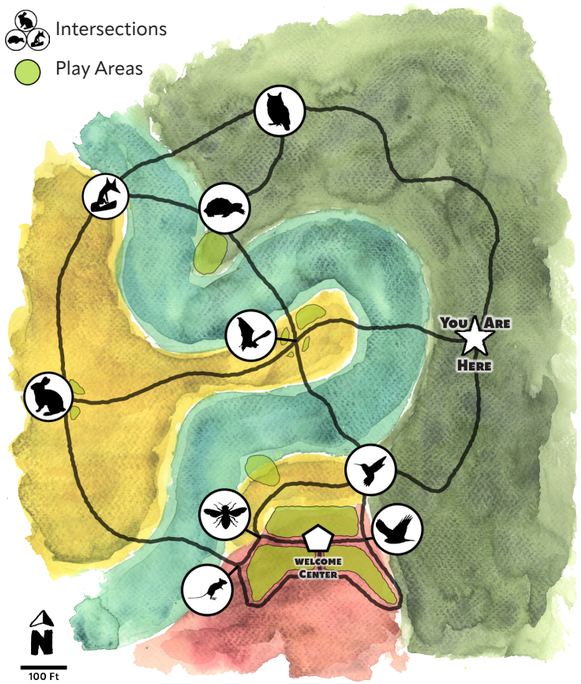

## Stormwater Systems



# **Trail Networks** | Suburban Ecotone Development | Ann Arbor, MI

To further cultivate a relationship between nature and humans, I designed the ecotone development with an expansive educational trail network that enables movement between landscape zones, and creates a pedestrian network connecting residents to vital commercial services.

## Example Trail Kiosk



### Eastern Grey Treefrog
*Hyla versicolor*

The eastern gray treefrog lives in the trees of hardwood forests in Southeast Michigan. They are experts in camouflage, and can change the color of their skin based on their surroundings. During the winter, these frogs hibernate in a semi-frozen state. As they thaw in the springtime, you can begin to hear their musical calls echoing throughout the forest at dawn and dusk.

https://nationalzoo.si.edu/animals/gray-tree-frog

## Trail Map and Welcome Sign

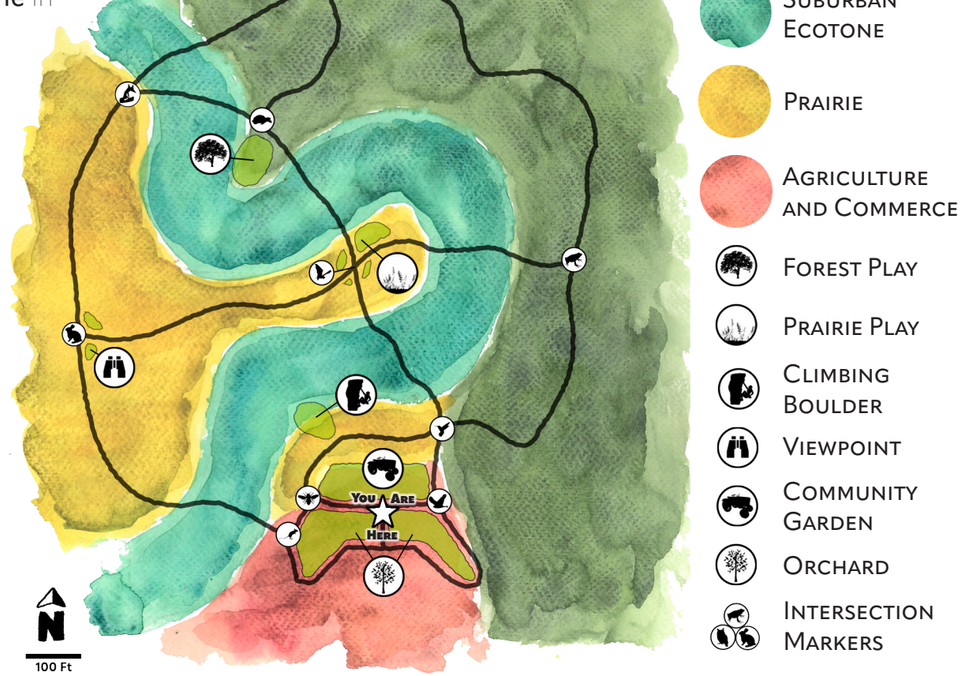
### Welcome to the Suburban Ecotone Park Network

An **ecotone** is an area of transition between ecosystem types. As you wander the trails, you'll transition from **prairie**, to **forest**, to **agricultural land**, and notice the **suburban ecotone** in between.

Play areas provide opportunities for all ages, and represent each unique ecosystem.

Kiosks at every intersection provide way-finding and information about each unique animal.

Throughout the park you'll find **adventure**, **play**, and **discovery.** Have fun exploring!



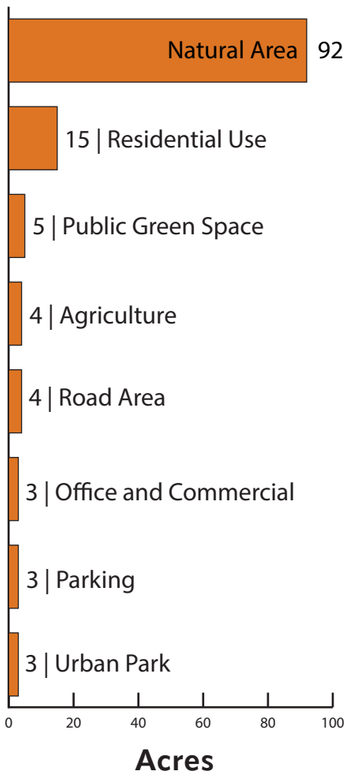
# **Conservation Design** | Coyote Development | York Township, MI

*Coyotes have lived alongside humans for millennia, surveying the edges of civilization in search of discarded resources, while captivating the edges of human experience through myth and story. Coyotes connect the wildness of natural systems to the order of human settlement, creating the Coyote Zone in between. This design prioritizes the Coyote Zone to facilitate connections between the natural and built environment, and to maximize the continuity of natural space.*

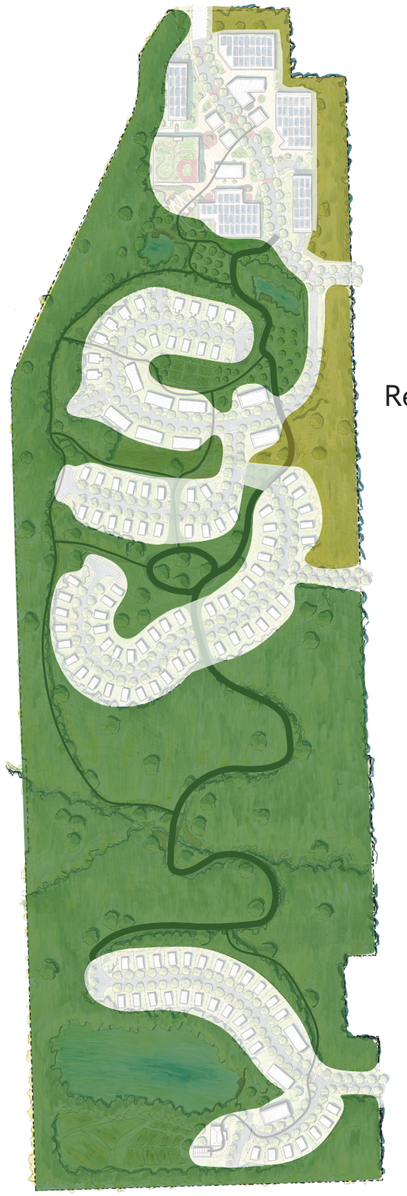
My design demonstrates coexistence between conservation and human development, as I not only conserved and enhanced a critical forest habitat but also met the infrastructure needs of the local community. For this design, I was tasked with developing a 118-acre site with 180 housing units and 240,000 sq-ft of office and commercial space. The site was located in York Township, MI, a region that has lost 93% of its historical forests, leaving a highly fragmented matrix. My design conserved 100% of the forest on site, added additional forest habitat through restoration, and prioritized contiguous habitat. Furthermore, to address the issues of regional habitat connectivity, I proposed a culvert enhancement to serve as a wildlife corridor across Highway 23, connecting the study site to the adjacent park.

## Ecological Matrix Context Map

## Planned Land Use Tabulations



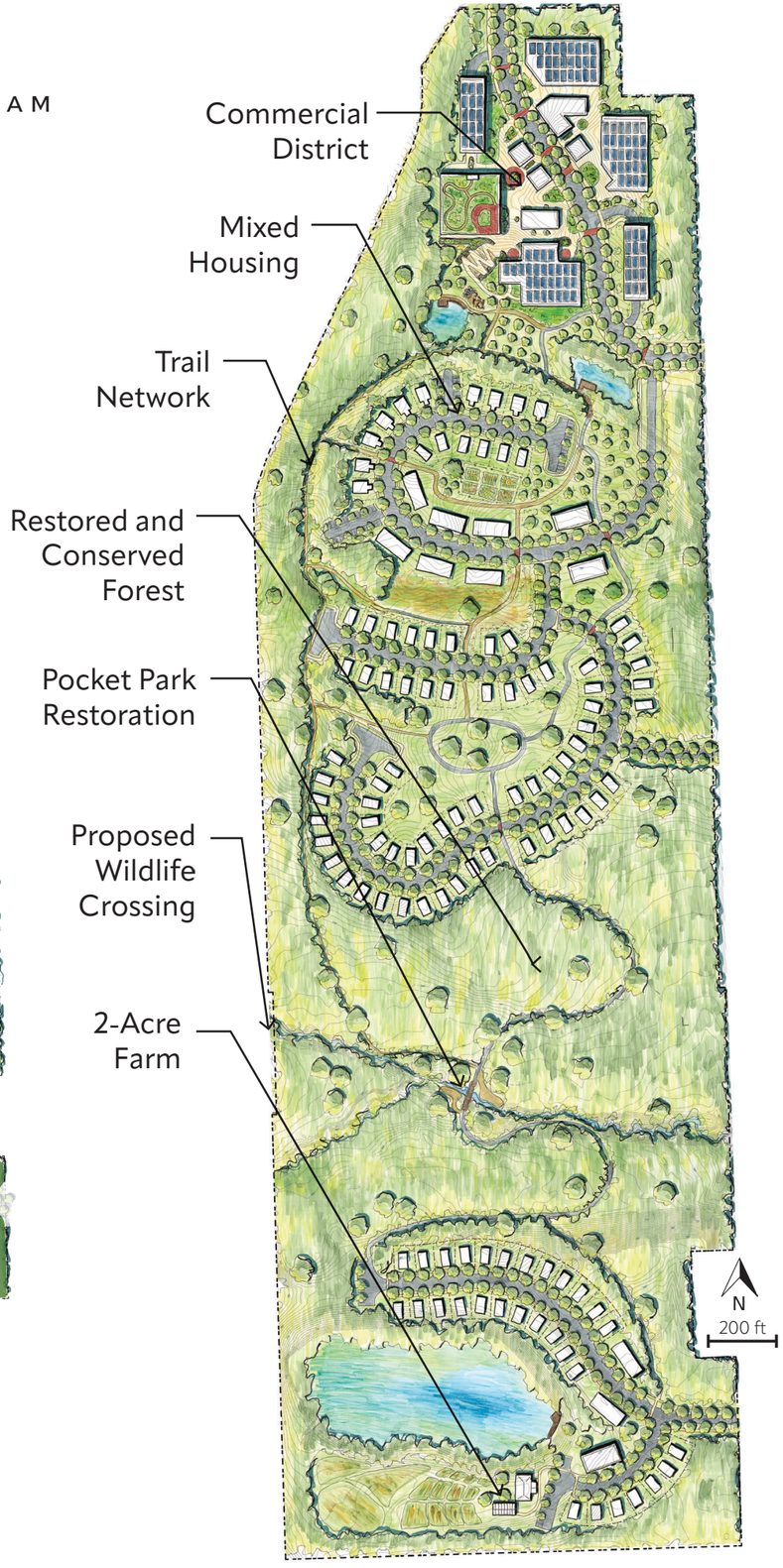

## Land Use Diagram



Conserved and Restored Habitat Patch

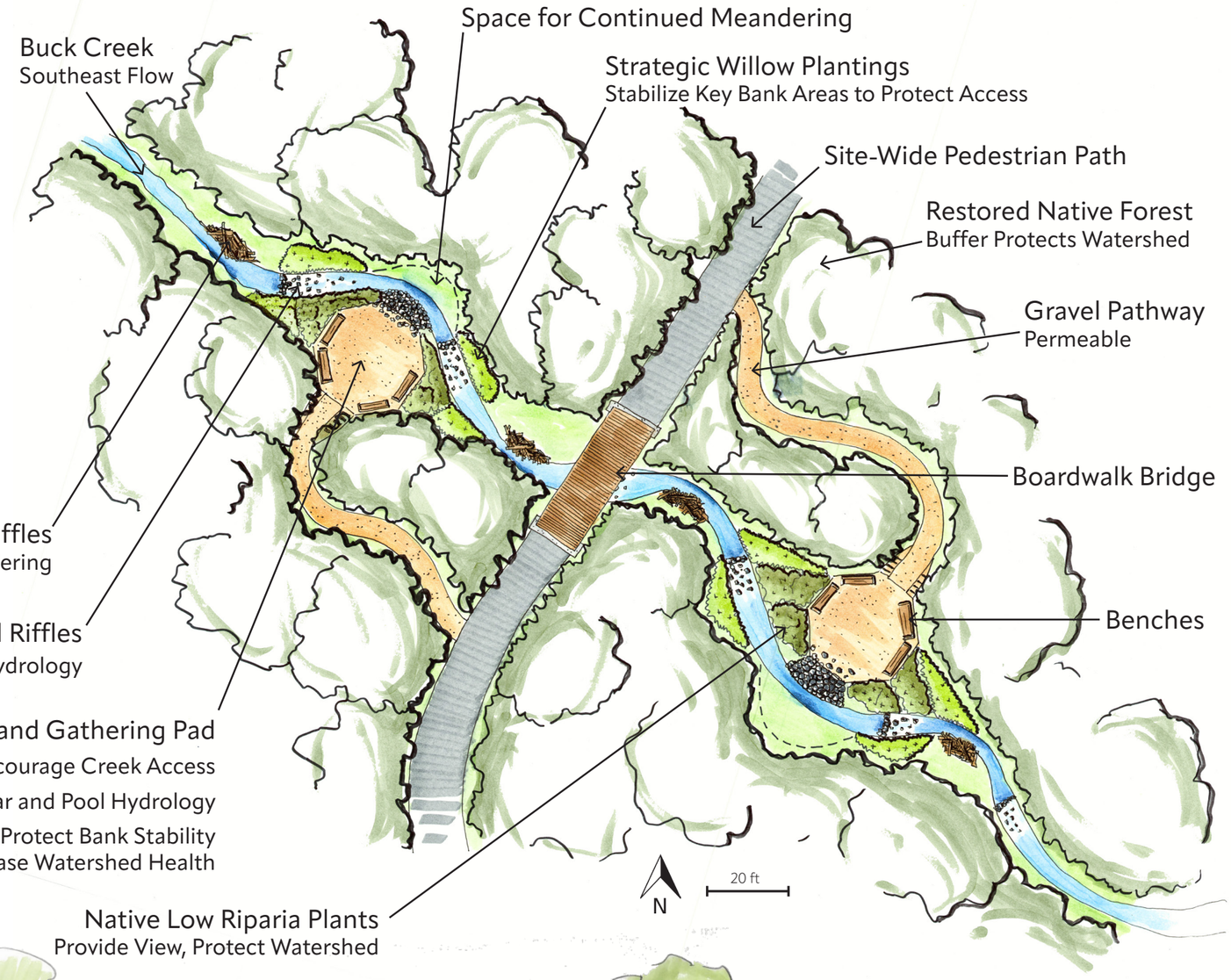
Vegetative Buffer

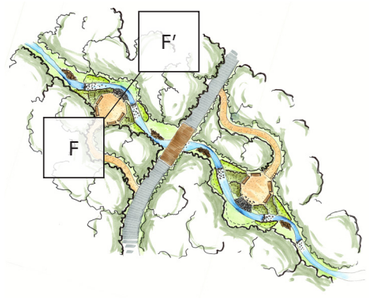
Development

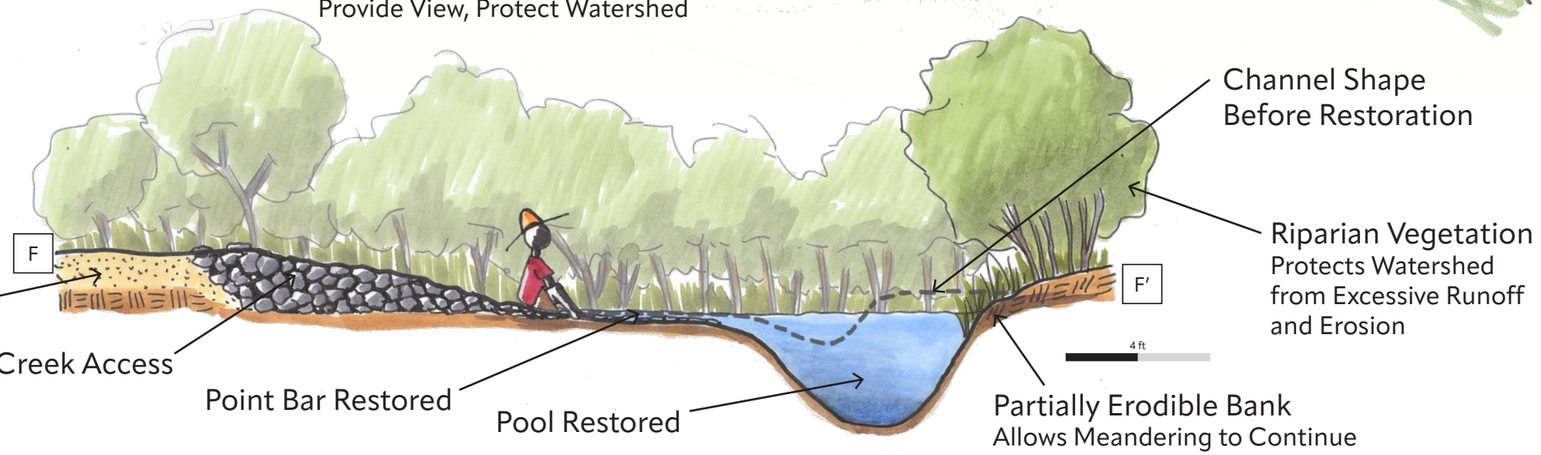


# **Restoration Design** | Coyote Development | York Township, MI

As part of the Coyote Development Project I designed a pocket park integrating creek restoration into the user experience of the water. Presently, the Buck Creek has a straight and eroded channel leading to poor habitat and water quality. To improve the hydrological condition of the stream, I designed a series of baffles and weirs to restore the natural structure of meanders, pools, and riffles. Together, these features increase habitat heterogeneity, water quality, and resilience to future disturbance. I used the natural point bar formation on meanders to provide durable access points for humans to enjoy the water without damaging the restored creek.



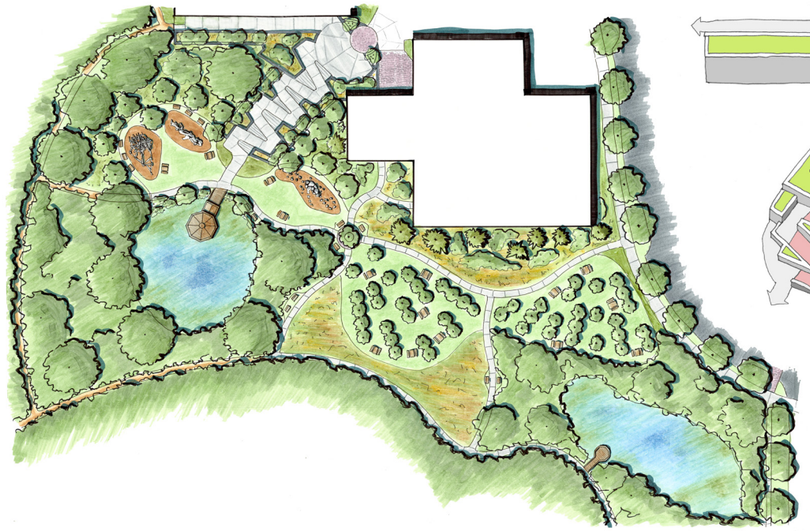



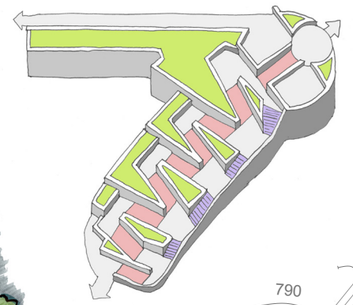
PRECEDENT IMAGE:
INDUCED MEANDERING WITH STICK AND POST BAFFLE, ABSAROKEE, MT

# Construction Drawings and Grading | Coyote Development | York Township, MI

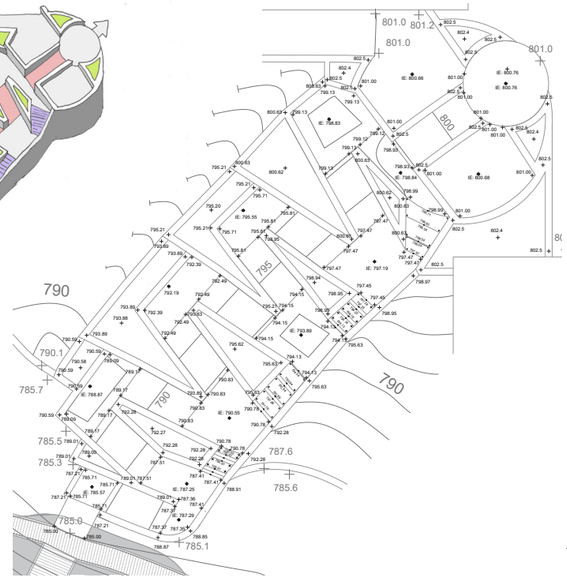
## Public Park Master Plan



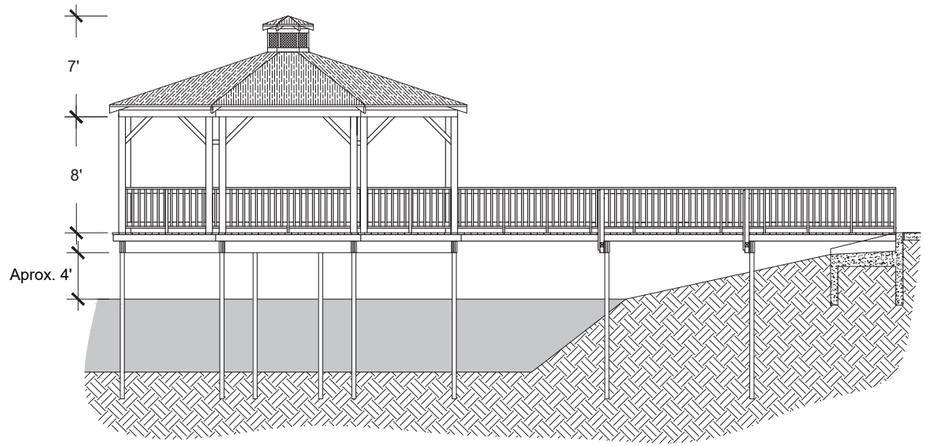
## "Stramp" Axon



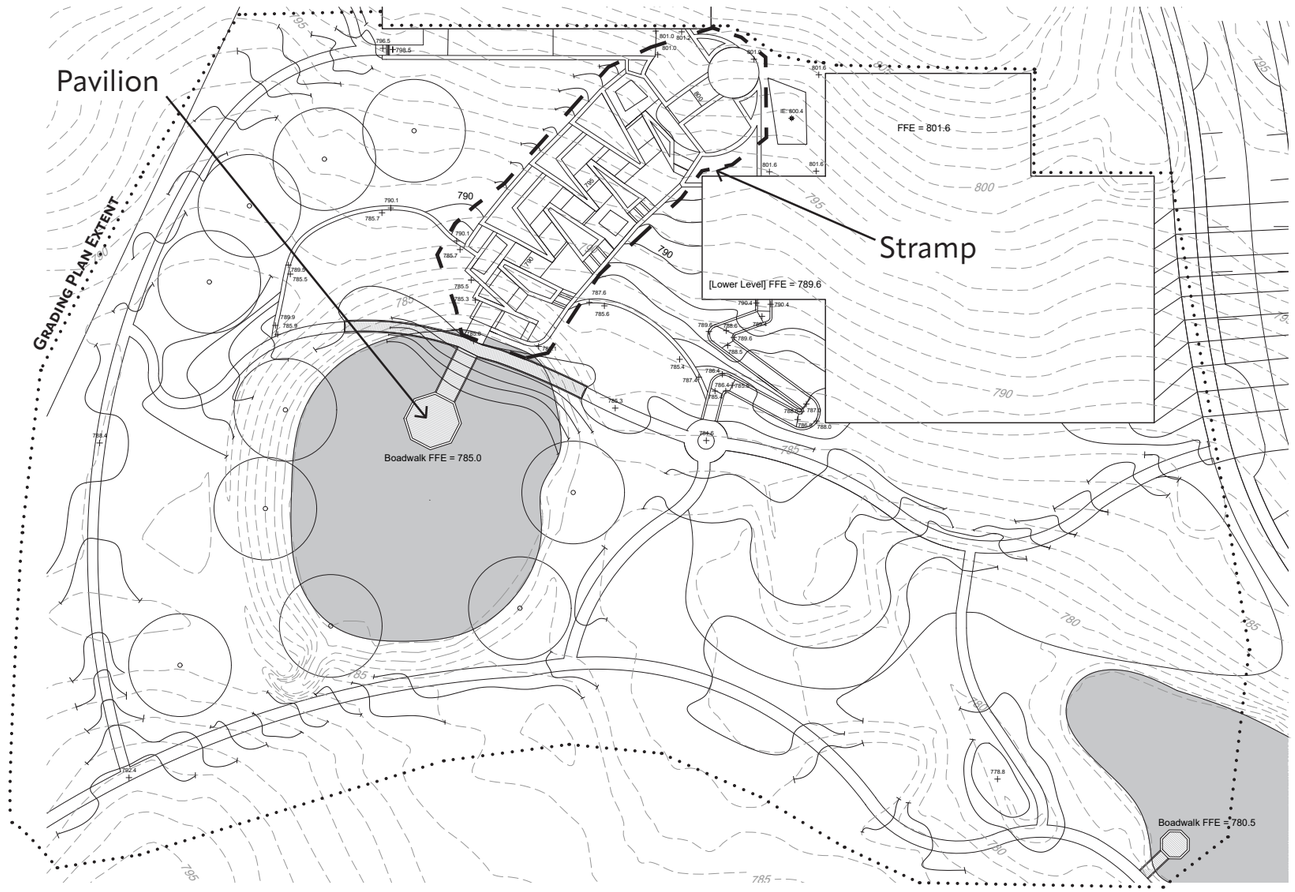
## "Stramp" Grading Detail




## Pavilion Elevation Detail

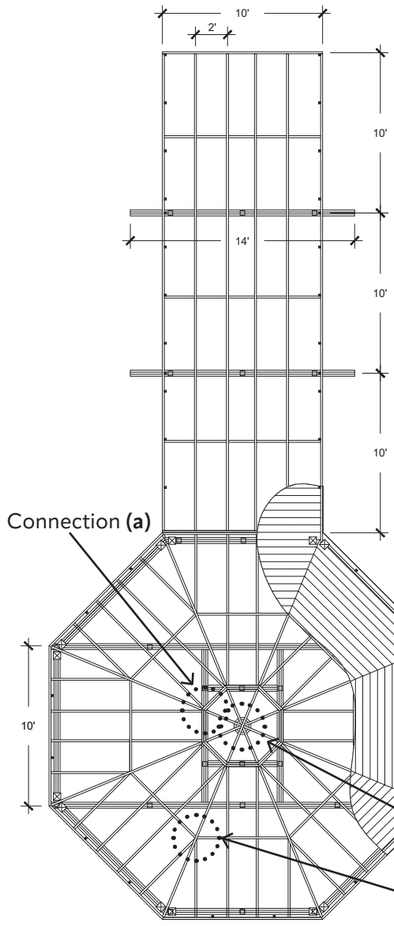



## Public Park Grading Plan

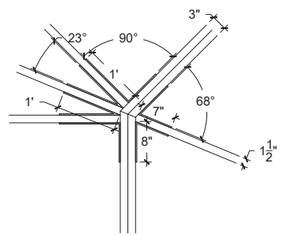



## Pavilion Plan Detail

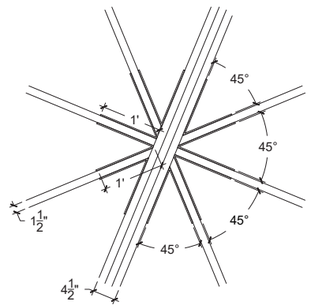
The following joist connections utilize custom fabricated joist hangers made of black powder coated steel. Angles and segment lengths noted in each connection detail. Each hanger is a single piece, with the top being open such that joists slot into the top.



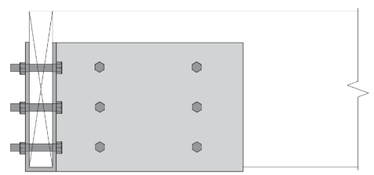

### Connection (a)



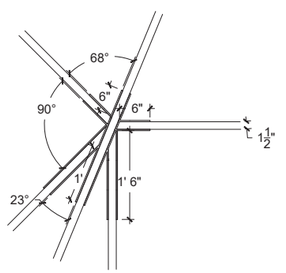

### Connection (b)




### Joist Hanger Section

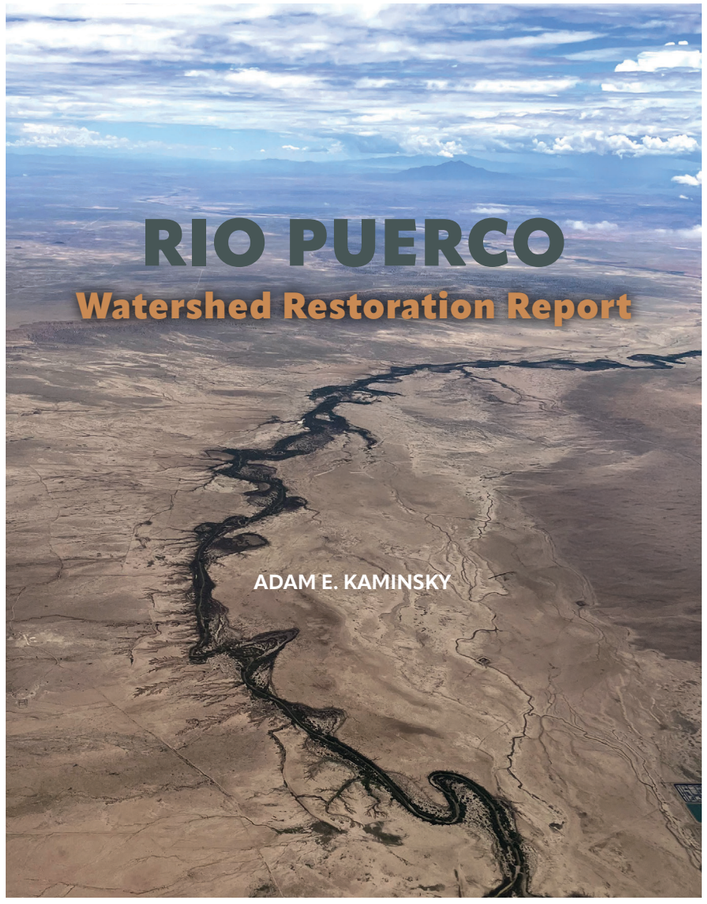


### Connection (c)



# Non-Profit Work | Rio Puerco Watershed Restoration | New Mexico

For my work as a Wyss Scholar, I developed a watershed restoration report for the Quivira Coalition as they investigated working in the Rio Puerco watershed in New Mexico. I researched, designed, and compiled a report that provided an overview of the history and current state of the land, identified important stakeholders, and recommended future actions. To the right is an example spread identifying the major landowners and potential collaborators for future work, and on the following page are selections from the report.



HEAVILY ERODED FENCELINE IN THE MIDDLE RIO PUERCO

**Jicarilla Apache Nation**

The Jicarilla Apache Nation is largely outside of the Rio Puerco watershed. However, they have excellent examples of traditional/ regenerative ranching elsewhere on their land, which could be brought into the watershed.
**facebook.com/jicarillaapachenationpr**

**Pueblo of Jemez**

The Jemez Pueblo has an organized rangeland management program, and members of the Pueblo actively practice traditional/ regenerative ranching.
**jemezpueblo.org/government/natural-resources**

**Eastern Chapters Navajo Nation**

The Torreon and Ojo Encino chapters of the Navajo Nation already have connections to Rio Puerco restoration. Many other chapters might be active in watershed restoration as well, but their participation is unknown. More information is needed to understand the role of these chapters in the watershed.

**Private Landowners**

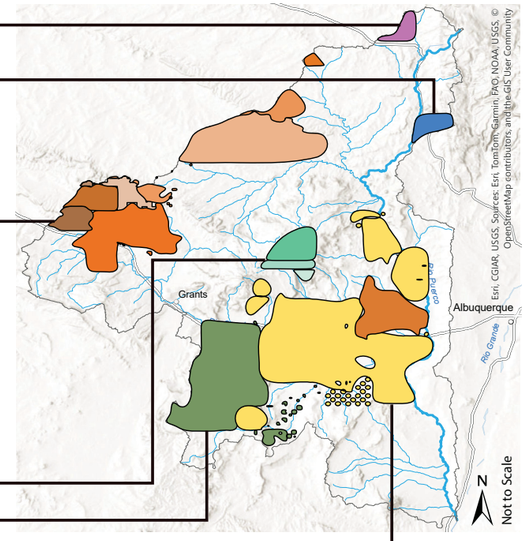
The plateau north of Mt. Taylor exhibits high grass and forb growth, giving it great potential for regenerative grazing. Much of this land is owned by 3 private corporate landowners: Fernandez Wind Energy (North), Fernandez Co Limited (Middle), and Lobo Partners (South). These companies might be interested in collaborating on Rio Puerco restoration.

**Acoma Pueblo**

The Acoma Pueblo manages a large tract of land in the Rio Puerco watershed, much of which exhibits high rangeland potential based on annual biomass data. Their Natural Resources Department manages Pueblo Rangeland.
**puebloofacoma.org**



## FUTURE COLLABORATIONS

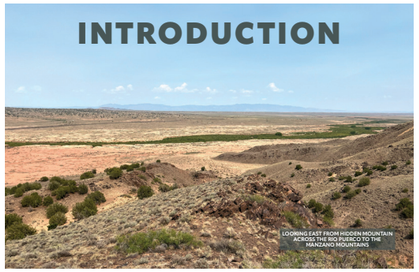


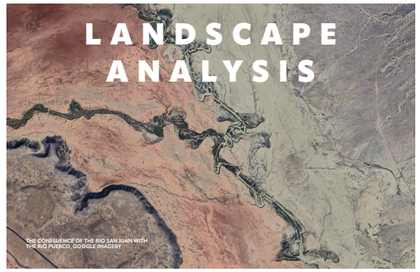

**Laguna Pueblo**

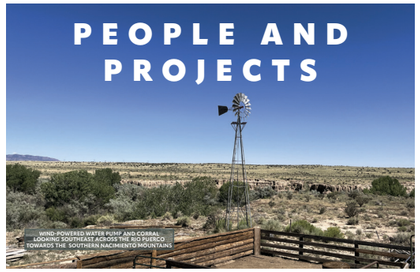
The Laguna Pueblo controls the largest tract of continuous Pueblo land in the Rio Puerco Watershed, which contains a long section adjacent to the Rio Puerco itself. Their Rangeland Management Program or Natural Resources Department might be interested in collaborating on Rio Puerco restoration.
**lagunapueblo-nsn.gov/departments/environmental-and-natural-resources/range-management-program**

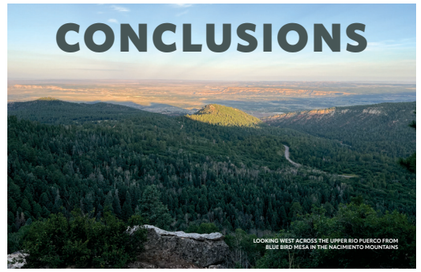
**Science**

Increased scientific engagement through universities and independent organizations could help practitioners and land stewards better understand and manage the watershed. Specifically, a growing body of research, led by an organization called CrustNet, demonstrates the efficacy of biological soil crusts as tools for restoration. Given the aridity and abundance of bare soil in the watershed, biocrusts could play a critical role in restoration, and CrustNet could help provide resources and expertise in this field.








## PRE-20TH CENTURY

Despite inconsistent water, the Rio Puerco watershed has a long history of human inhabitance. Within the watershed, over 10,000 cultural sites have been identified, the oldest of which are dated to 9500 BCE (Curt Larsen and Martha Herzog, 2000). In ancient times the Chacoan people, and later the Ancestral Puebloan people, heavily utilized the region, and their descendants continue to occupy the watershed. In the southern reach of the river, the Pottery Mound Pueblo sat along the banks of the Rio Puerco from 1300-1550 CE, and in modern times is famous for its beautifully painted Kiva murals and finely made ceramics (Franklin, 2018). In more recent history, numerous Indigenous groups have inhabited the region, utilizing the landscape for wild game, medicinal and edible plants, and farming corn, chiles, squash, melons and turkeys (Bryan, 1928; Pueblo of Isleta, 2023; Roger Fragua, personal communication, July 18, 2025). Today, the Acoma, Isleta, Jemez, and Laguna Pueblos, 14 chapters of the Navajo Nation, and the Jicarilla Apache Nation all officially reside within the bounds of the Rio Puerco watershed. Many other Indigenous groups are located nearby.

"Mosquito Man", Kiva Mural Painting from Pottery Mound (Vanpool & Velandia Jagua, 2022).

Prior to the 1800s, the Rio Puerco existed as a small channel with gently sloping sides that frequently flooded the surrounding valley floor (Bryan, 1928). In the 1760s, the Spanish colonial governor approved grazing grants in the watershed, prompting hopeful colonists to move into the region with the intention of grazing livestock. Despite a modest attempt at settlement, Navajo and Apache raiders defending their territory ultimately chased the colonists back to Albuquerque, leaving the valley minimally impacted by livestock (Diggle & Hieb, 2004). In 1868 a treaty was passed in which the Navajo Nation agreed to cease fighting with the United States in return for official rights to their homeland, largely situated outside of the Rio Puerco watershed (Native Knowledge 360, 2019). With the passing of this treaty, colonists were free to bring large herds of livestock into the valley.

Sheep grazing in the Rio Puerco Valley, 1880s (Rocky Mountain Research Station, 1998).



## 20TH CENTURY

The valley in the late 1800s was ecologically rich and provided excellent rangeland for livestock. Periodic flooding replenished the land with nutrients and water, and the river was easily dammed and diverted through acequias to irrigate crops. By the end of the 19th century, settlers had thoroughly capitalized on these resources by establishing small towns, building an extensive network of dams and acequias, and grazing vast herds of livestock on both sides of the Rio Puerco (Diggle & Hieb, 2004). News of prosperity in the valley spread, and the Rio Puerco became known as the "breadbasket of New Mexico."

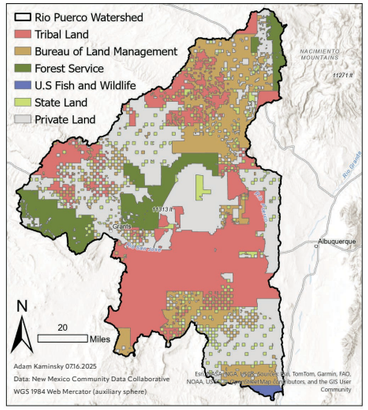
Abandoned structure in the historic town of Guadalupe.

Geography of the Middle Rio Puerco Valley, 1959 (Gwayn Widdison, 1959).

However, this success was relatively short-lived. As early as 1890, the rangelands of the middle valley were reported to be overstocked, and the vegetative cover was quickly eaten away by dense herds of livestock (Gwayn Widdison, 1959). Without the protective barrier of plants, the loose soils of the valley became vulnerable, and accelerated erosion began to dramatically change the shape of the valley. The river channel began to deeply incise, lowering the water table and cutting the river off from its historic floodplain (Diggle & Hieb, 2004). As deep incisions migrated upriver, they dewatered the surrounding valley until what was once a richly vegetated floodplain became sere uplands with little resources to support life. Of the many towns that were established in this brief period of prosperity, many were abandoned, and those that remained became steadily more depopulated and impoverished alongside the land (Bryan, 1928). Nonetheless, the Rio Puerco valley was never abandoned, and today many communities continue to call this watershed home.

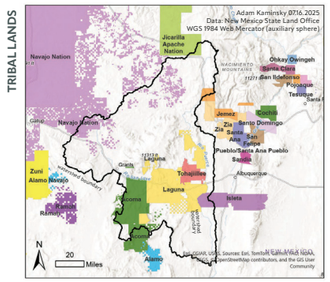


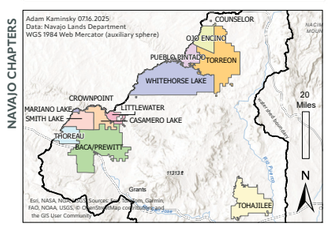
## LAND OWNERSHIP




LEGAL LAND OWNERSHIP IN THE RIO PUERCO WATERSHED





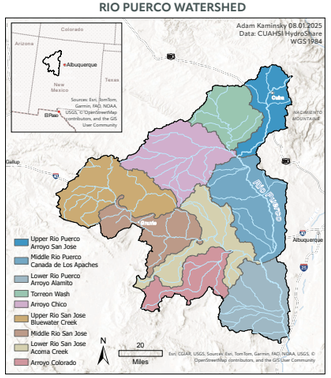






## GEOGRAPHY OF THE REGION

The Rio Puerco is situated in northwestern New Mexico, where it runs 150 miles south from the Nacimiento Mountains to the Rio Grande. The first 10 miles flows perennially, while the rest is ephemeral.

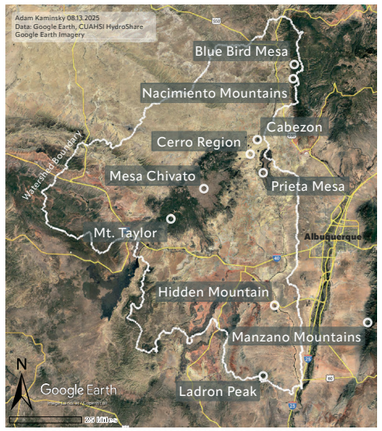
RIO PUERCO WATERSHED






## LANDMARKS

This map displays various landmarks noted in photographs throughout the report. Names reflect the common, colloquial naming of the region, and do not reference the numerous traditional Indigenous place names for these landmarks.




COLLOQUIAL LANDMARK NAMES IN THE RIO PUERCO WATERSHED



## RECOMMENDATIONS

This report identifies the need for collaborative restoration in the Rio Puerco watershed, and recommends the following interventions:

### 1: Improve habitat and control erosion within the Rio Puerco and surrounding upland ecosystems.

To achieve this goal, people and organizations will need to actively pursue restoration projects in the watershed. A substantial body of literature and a rich traditional ecological tradition describe numerous methods of controlling erosion and restoring upland, riparian, and aquatic ecosystems. However, time and resources are common roadblocks to enacting these methodologies.

This report provides information and data supporting the need for restoration, and the findings of this report may be included in grants to potential project funders. The report also indicates areas of focus for future work, and identifies the major organizations with connections to the watershed.

### 2: Foster a community of practice that communicates regionally in pursuit of innovative land management and erosion control strategies.

To create a community of practice around restoration, it is first necessary to identify potential community members. This report identifies many people and organizations with connections to the land, and therefore a potential stake in restoration efforts. Two communities stand out as potential starting points for a broader community of practice:

The **Ojo Encino Ranching Committee** and **Ranchers in the Torreon Chapter of the Navajo Nation** are currently involved in restoring the watershed in collaboration with **River Source** and **Rio Grande Return**.

The **Comanche Ranch** in the **Isleta Pueblo** has a history of rangeland restoration and erosion control, and may be interested in rejuvenating restoration efforts.

From these starting places, a broader community of practice could be established that incorporates the people and organizations listed on the "Future Collaborations" page, with a regional focus on sustainable grazing practices in areas with high rangeland potential.



### 3: Increase collaboration among relevant organizations across the fields of agriculture, conservation, and restoration.

Collaborative watershed restoration is enriched by expanding to a broad community of farmers, ranchers, scientists, planners, and practitioners. This report identifies that the **BLM, Rio Grande Return, and River Source (with a longtime independent contractor named Watson Castillo)** are currently contributing to restoration in the watershed at this time. However, it is likely that numerous Tribal Governments are actively working towards conservation and restoration as well.

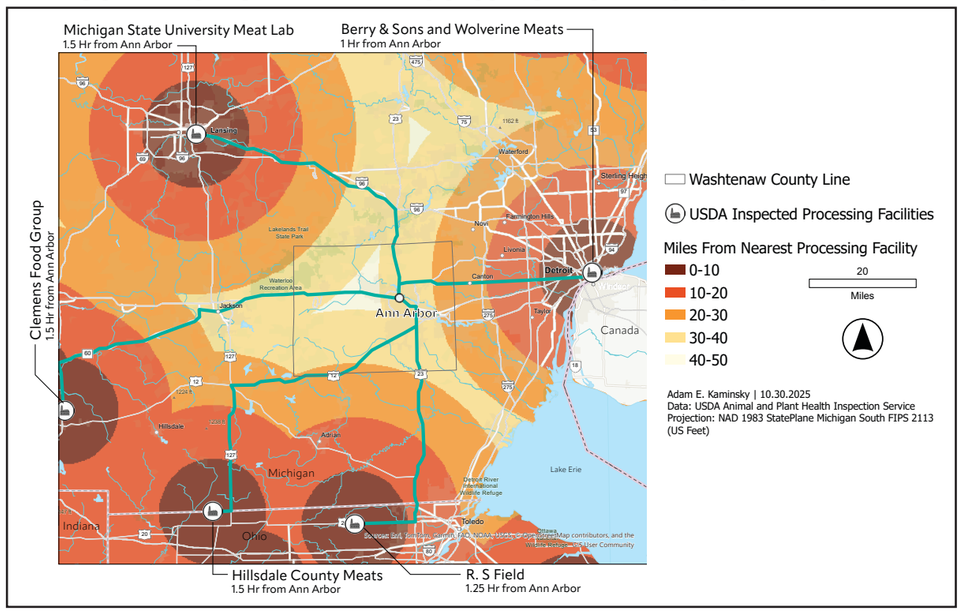
Restoration planners and practitioners currently dominate the watershed restoration effort, so this report recommends diversifying collaboration to include farmers, ranchers, and scientists via **Tribal Governments**, **ranchers utilizing BLM leases**, **private land owners**, **CrustNet** and **New Mexico colleges and universities**.

Cerro de Guadalupe at sunrise with the Rio Puerco just behind.
Wildfire smoke streaming over Mesa Chivato in the background.

CONFLUENCE OF THE RIO PUERCO WITH THE RIO GRANDE, SOCORRO COUNTY, NM

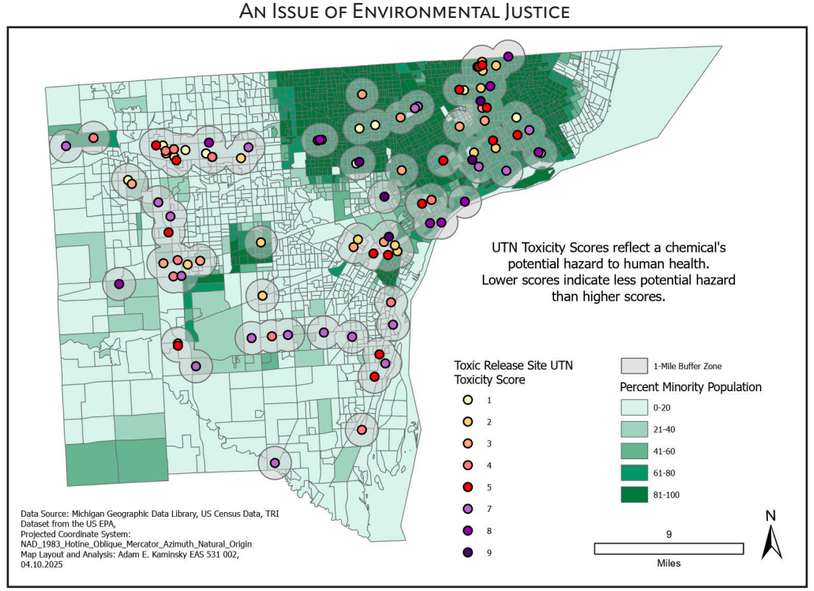
# Geospatial Data Analysis | Various Projects

## USDA Inspected Meat Processing Facilities: Southeast MI



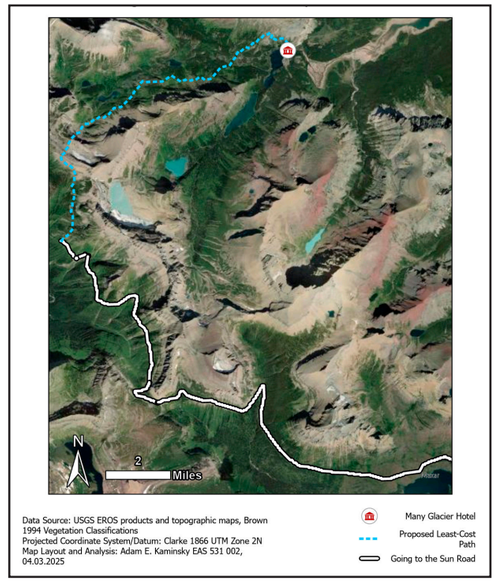

## Toxic Release Sites: Wayne County, MI

### An Issue of Environmental Justice



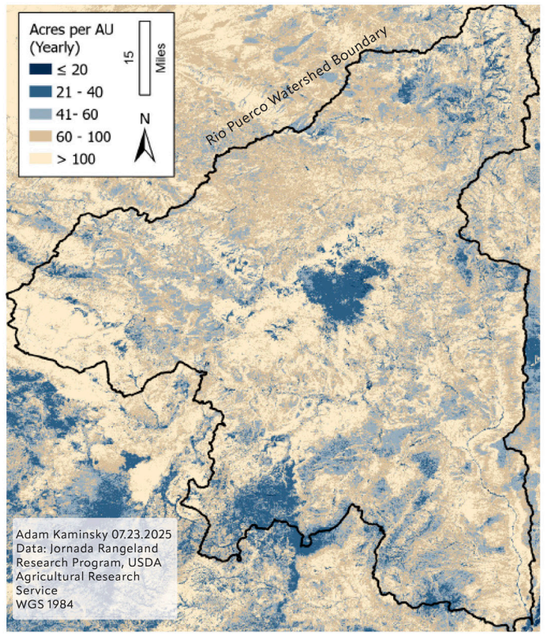

## Least-Cost Path Proposal: Glacier National Park, MT

### Going to the Sun Road to Many Glacier Hotel




## Rangeland Potential: Rio Puerco, NM

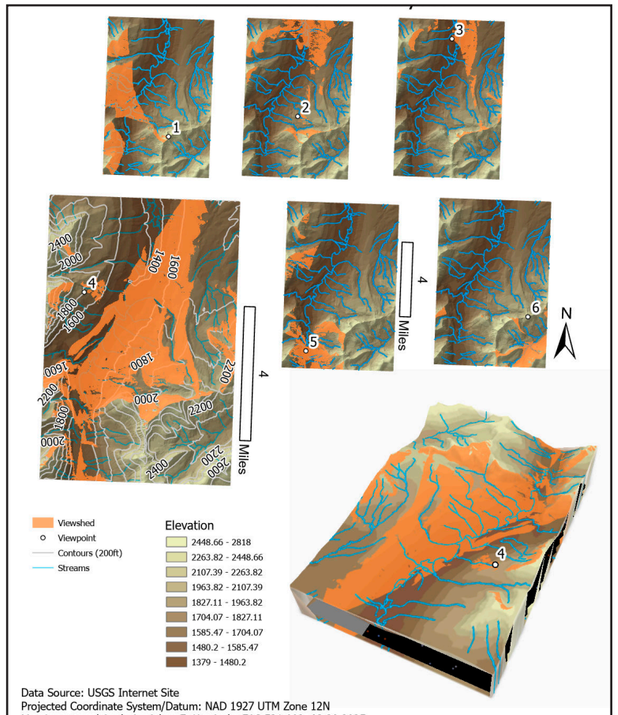
### Based On Annual Forage Biomass



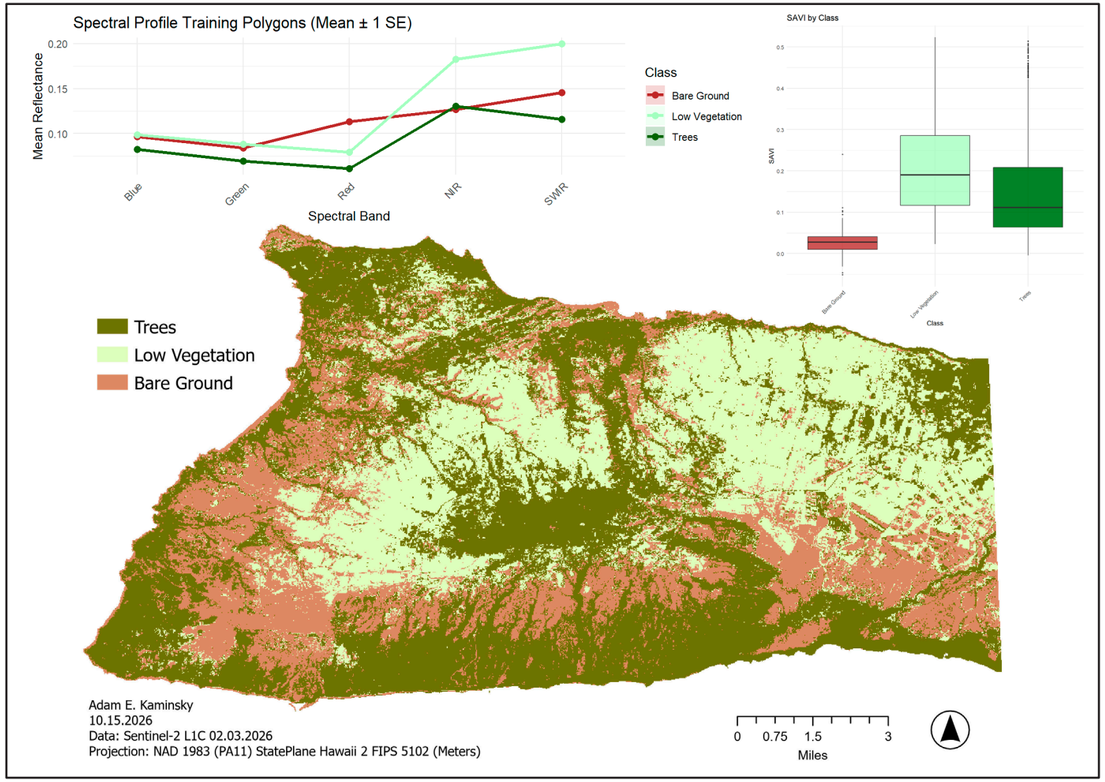

On average, ranchers in this region of New Mexico need 40-60 acres to support one animal unit (AU). Therefore, viable rangeland is represented by areas with <60 acres/AU.

## Terrain Analysis: Gable Mountain, MT

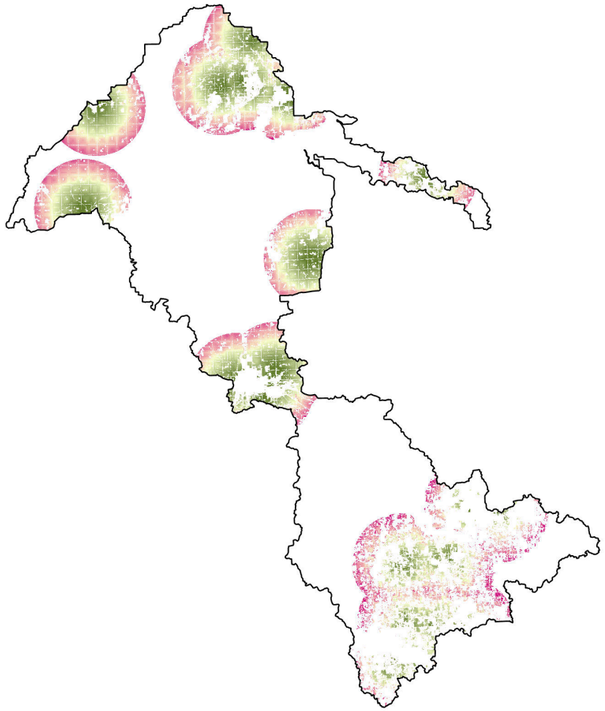
### TIN and Viewshed Analysis



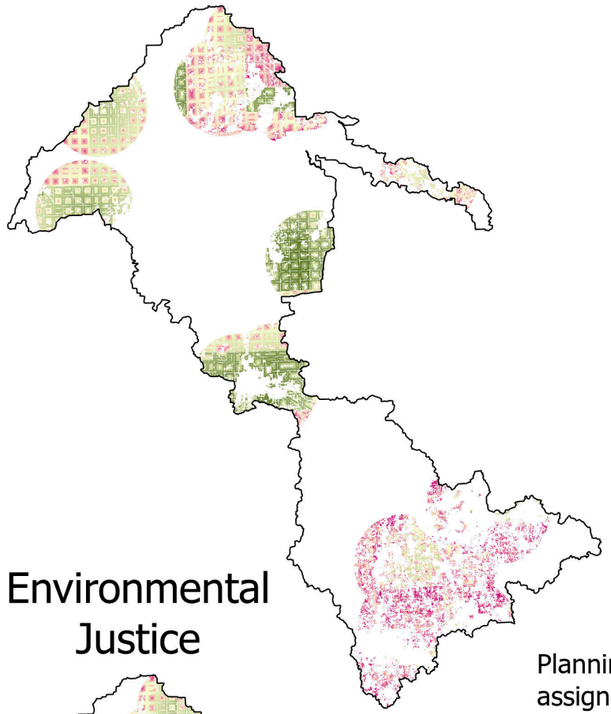

## Remotely Sensed Land Cover Classification: Molokai, HI



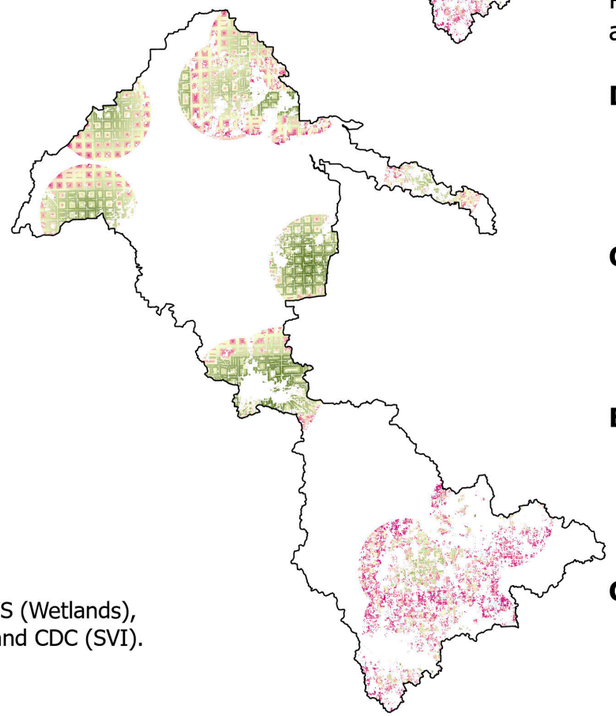
Developer-Led



Community First



Environmental Justice



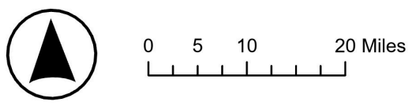
**Planning Scenarios**

Developement Suitability

10 (Most Suitable)

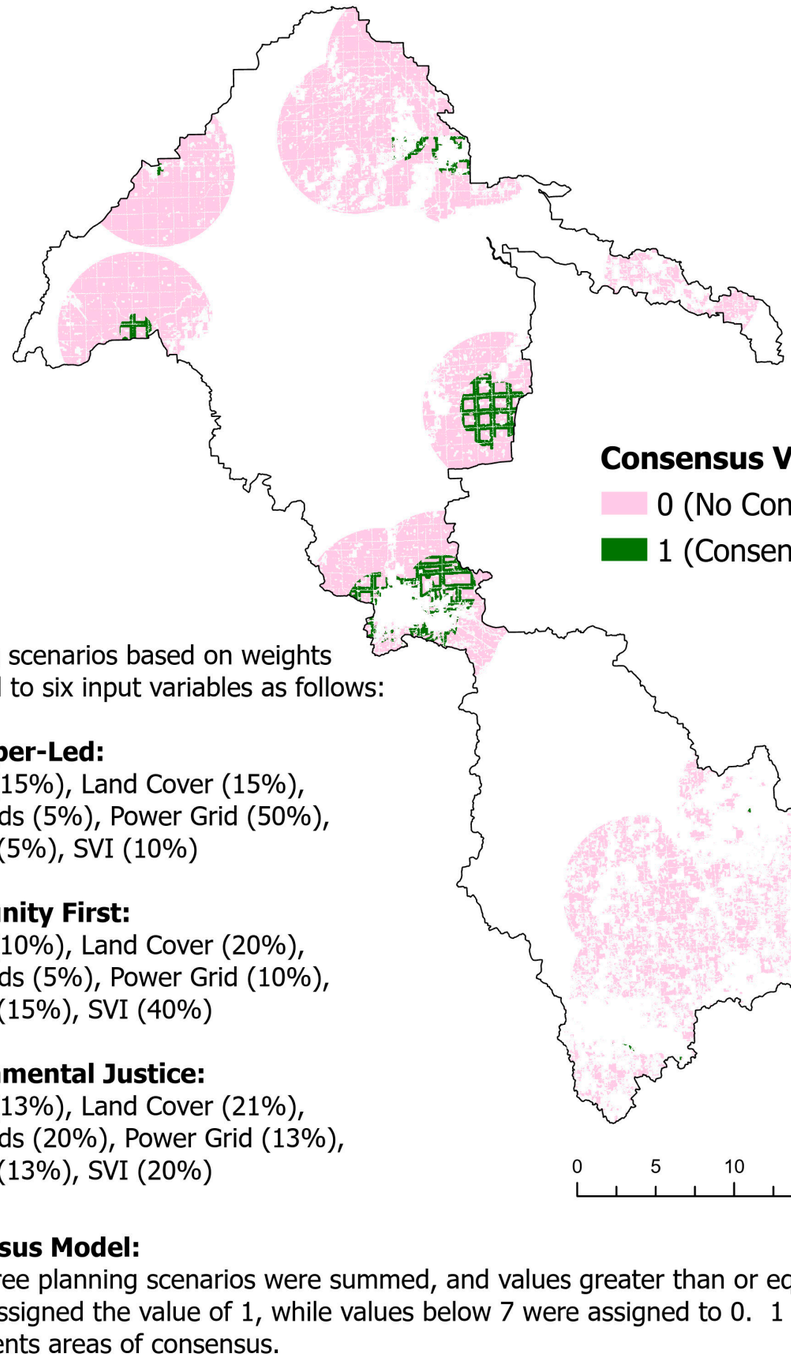
1 (Least Suitable)

Shiawassee Watershed Boundary




Adam E. Kaminsky
02.19.2026
Data: USGS (Slope), NLCD 2024 (Land Cover), USFWS (Wetlands), ArcGIS Online Library (Power Grid), MDOT (Roads), and CDC (SVI).
Projection: NAD 1983 Michigan GeoRef (Meters)

**Consensus Model**



**Consensus Value**

0 (No Consensus)

1 (Consensus)

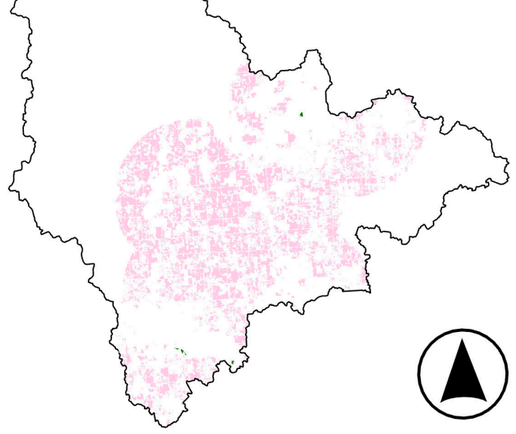
Planning scenarios based on weights assigned to six input variables as follows:

**Developer-Led:**
Slope (15%), Land Cover (15%), Wetlands (5%), Power Grid (50%), Roads (5%), SVI (10%)

**Community First:**
Slope (10%), Land Cover (20%), Wetlands (5%), Power Grid (10%), Roads (15%), SVI (40%)

**Environmental Justice:**
Slope (13%), Land Cover (21%), Wetlands (20%), Power Grid (13%), Roads (13%), SVI (20%)

**Consensus Model:**
The three planning scenarios were summed, and values greater than or equal to 7 were assigned the value of 1, while values below 7 were assigned to 0. 1 represents areas of consensus.



## Creek Restoration

Condition Before:
Deeply eroded channel

Condition After:
Rock Rundown and Splash Pool

Condition Before:
Straight and eroding channel

Condition After:
Woven Baffle / Beaver Dam Analogue

## Post-Fire Restoration

Condition Before:
Heavily Eroded Following Fire

Condition After:
Log Step-Down with Stone Base

## Regenerative Grazing

Grassland restoration with rotationally grazed sheep and cattle

## Human Infrastructure

Bridge construction to enhance creek crossing and lessen impact on creek edge

COTTONWOOD BOSQUE
ALONG THE JUDITH RIVER, MT